13)Implement Your own ping program

Program:

```
  GNU nano 8.7.1
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>
#include <sys/time.h>
#include <netinet/ip.h>
#include <netinet/ip_icmp.h>

unsigned short checksum(unsigned short *data, int len)
{
    unsigned long sum = 0;
    while (len > 1) {
        sum += *data++;
        len -= 2;
    }
    if (len == 1)
        sum += *(unsigned char *)data;
    sum = (sum >> 16) + (sum & 0xffff);
    sum += (sum >> 16);
    return (unsigned short)~sum;
}

int main()
{
    int sockfd;
    char buffer[1024] = {0};
    char *ip = "142.251.221.110";

    sockfd = socket(AF_INET, SOCK_RAW, IPPROTO_ICMP);
    if (sockfd < 0) {
        perror("socket");
        return 1;
    }

    struct timeval tv = {3, 0};
    setsockopt(sockfd, SOL_SOCKET, SO_RCVTIMEO, &tv, sizeof(tv));

    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    inet_pton(AF_INET, ip, &addr.sin_addr);

    struct icmphdr *icmp = (struct icmphdr *)buffer;
    icmp->type = ICMP_ECHO;
    icmp->code = 0;
    icmp->un.echo.id = htons((unsigned short)getpid());
    icmp->un.echo.sequence = htons(1);
    icmp->checksum = 0;
    icmp->checksum = checksum((unsigned short *)icmp,
                              sizeof(struct icmphdr));

    printf("Pinging %s...\n", ip);

    if (sendto(sockfd, buffer, sizeof(struct icmphdr), 0,
               (struct sockaddr *)&addr, sizeof(addr)) < 0) {
        perror("sendto");
        close(sockfd);
        return 1;
    }

    struct sockaddr_in from;
    socklen_t fromlen = sizeof(from);
    int n = recvfrom(sockfd, buffer, sizeof(buffer), 0,
                     (struct sockaddr *)&from, &fromlen);

    if (n >= (int)sizeof(struct iphdr) + (int)sizeof(struct icmphdr)) {
        struct iphdr *iph = (struct iphdr *)buffer;
        struct icmphdr *reply =
            (struct icmphdr *)(buffer + iph->ihl * 4);

        if (reply->type == ICMP_ECHOREPLY &&
            from.sin_addr.s_addr == addr.sin_addr.s_addr)
            printf("Reply received from %s\n", ip);
        else
            printf("Received a packet, but it was not the expected reply.\n");
    } else {
        printf("Request timed out or packet was too short.\n");
    }

    close(sockfd);
    return 0;
}
```

Output :

```
mikey@mikey:~/exp12$ gcc ping.c -o ping
mikey@mikey:~/exp12$ sudo ./ping
Pinging 142.251.221.110...
Reply received from 142.251.221.110
mikey@mikey:~/exp12$
```